



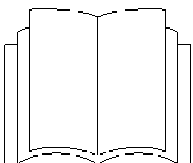
Operator's manual for Attachment



Sand Spreader

REAR MOUNT

1280 mm Rear mount Sand spreader for 300-700-series loaders: A21242



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

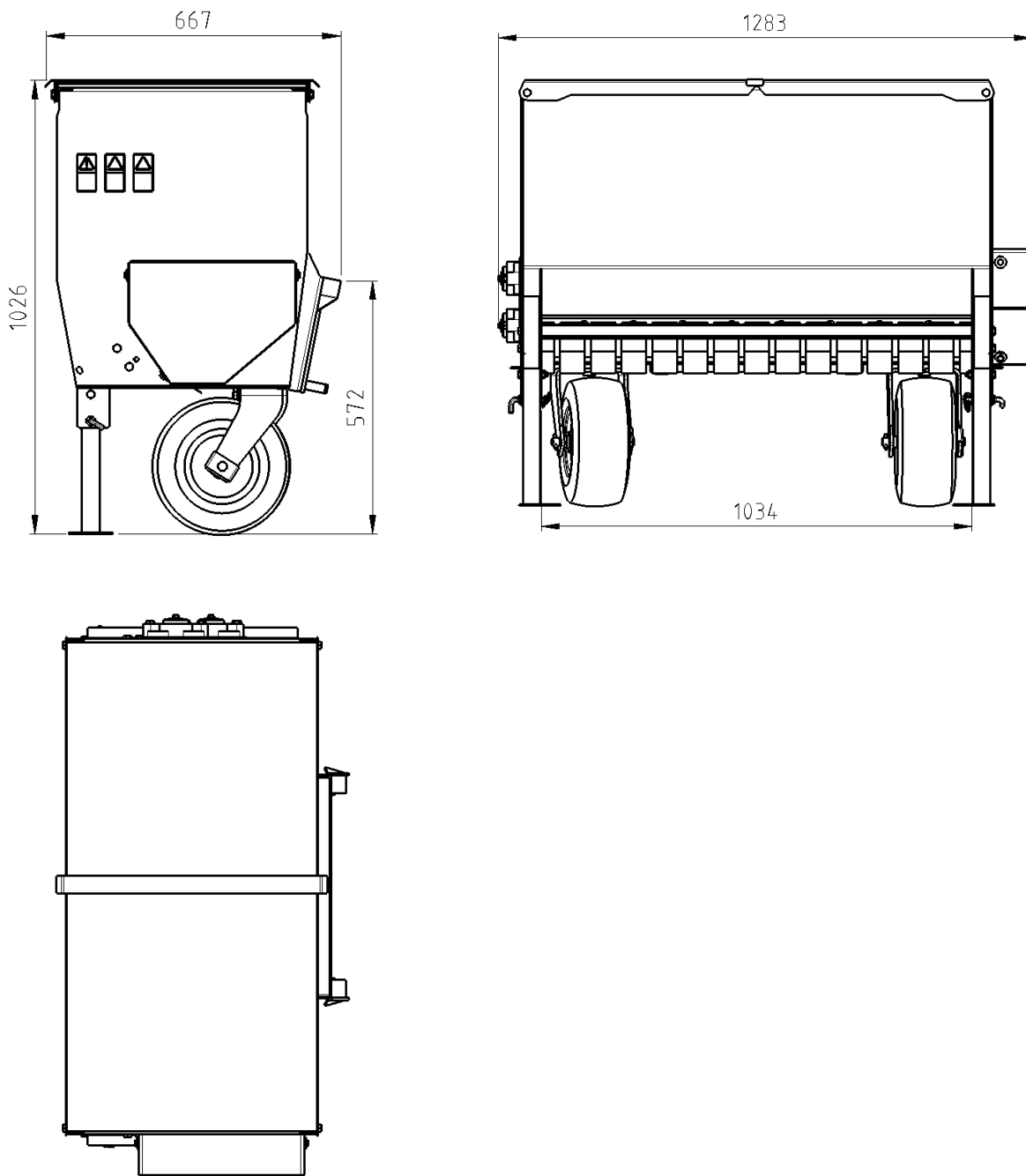
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Dimensions, Sand spreader A21242:



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Appendix: EC Declaration of Conformity

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of experience and product development. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

The purpose of this manual is to help you to:

- operate the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life

The following warning symbols are used throughout this manual to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	WARNING:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.

NOTICE

This notice mark indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader and attachment and all safety instructions before using the equipment. Using the equipment for other purposes or use in any other way than described in this manual is prohibited. Keep this manual at hand throughout the service life of the equipment. If you sell or transfer the equipment, be sure to hand over this manual to the new owner. If the manual is lost or damaged, you can request a new one from the retailer or manufacturer. Due to continuous product development, some of the details in this manual may differ from your equipment. Original instructions.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Particularly the regulations concerning the use of the equipment on public road areas must be observed. We reserve the right to change the contents of the manual without notification.

2. Designed purpose of use

This AVANT Sand/salt spreader is an attachment that has been designed and manufactured to be used with AVANT multi purpose loaders. The spreader is designed to be mounted on the back of the loader models as listed in Table 1, but can be also used normally when mounted to the quick attach system on the loader boom. The spreader is intended for fast and efficient spreading of sanding gravel, crushed rock, salt or other small fragmented material suitable for sanding.

The spreader can be conveniently mounted on the rear mount stand or rear hitch, making it possible to mount another attachment to the loader boom, such as a bucket or a dozer blade. This makes professional property maintenance more efficient. The spreader is operated by a hydraulic motor and is equipped with a mixing shaft that prevents clogging of material. When mounted on the loader boom, it is easy to use the spreader accurately even at tight areas.

Mounting the spreader to the back of the loader makes it necessary to have double acting rear hydraulics outlet on the back of the loader and a rear hitch device or rear mount bracket. Because the weight added to back of the loader by the spreader, the front of the loader may become too light and the front wheels may lift off the ground. To limit tilting of the loader, the spreader is equipped with stabilizing wheels. When using a rear mounted spreader, a bucket or heavier attachment must be mounted on the loader boom to maintain the balance of the loader. If the loader is also equipped with rear counterweight, it may not be possible to fill the spreader to its maximum capacity.

The attachment has been designed to require as little maintenance as possible. The operator can perform regular maintenance tasks. However, all repair work can't be performed by the operator, and demanding repair and maintenance operations, such as servicing or replacing hydraulic components, must be left for professional service technicians. All maintenance work must be done using proper safety equipment. Spare parts must be similar to original specifications, which can be ensured by using only original Avant spare parts.

Familiarise yourself with the manual's instructions regarding service and maintenance. Please contact your AVANT dealer if you have additional questions about the operation or maintenance of the equipment, or if you require spare parts or maintenance services.

Table 1 –Sand spreader A21242 compatibility with Avant loaders

Model	216 218 220	313S 320S 320S+	419 420 520	525LPG R28 R35	528 630	635 640	745 750
Front mount to quick attach plate of loader boom	-	-	-	•	•	•	•
Rear mount to rear lift system or mounting bracket	-	-	-	-	(•)	•	•



When mounting the spreader to the back of the loader, a sufficiently heavy attachment must be mounted to the front to maintain balance of the loader. Rear counterweights can't be used.

3. Safety instructions for using the sand spreader

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards.



Misuse, careless use, or using an attachment that is in bad condition may cause serious injuries. Familiarise yourself with the controls of the loader and coupling and safe operation of the attachment at a safe area. Become familiar especially with how to stop the equipment in a safe manner. Read the following safety precautions carefully.

Read all safety instructions carefully before handling the attachment



- When attaching the attachment to the loader, **ensure that the locking pins of the loader's quick attach plate are positively in the lower position and that they have locked the attachment to the loader.**
- The Sand Spreader is to be used by one person at a time. Do not go near a running attachment or allow others to approach the area where the machine is in operation.
- Never leave the driver's seat when the spreader is running or when the auxiliary hydraulics control lever of the loader is locked on. Keep a **safety distance of at least one metre** between activated sand spreader and any person in vicinity.
- Make sure that an uncoupled attachment is properly supported. Unmount the spreader carefully and use the support stands of the spreader. Ensure that the spreader is adequately supported during all maintenance, service, or adjustment tasks.
- Keep hands and feet away from moving parts. Beware of entanglement hazard of a running spreader. Any inspection, cleaning, or clearing of blockages must be done carefully. Always stop the attachment following the safe stopping procedure before reaching inside the spreader.
- Stop the spreader and shut down loader engine when filling the spreader.
- Always transport the spreader as low and close to the machine as possible to keep the centre of gravity low and for the best stability. On slippery areas drive calmly and deliberately. Operate the controls of the loader slowly and smoothly especially on inclined surfaces.
- If mounted on the loader boom or rear lift device, lower the attachment firmly on the ground before leaving the driver's seat. Make sure that the attachment is properly supported during maintenance or inspections. Set the outriggers of the attachment always when unmounting the attachment from the loader. Never leave the driver's seat when the attachment is lifted. Going under a raised attachment or loader boom is dangerous, as the boom may lower due to loss of stability, mechanical fault or by other person operating the controls of the loader.
- Follow safe stopping procedure before any maintenance or adjustments. Before clearing blockages, the attachment must always be stopped following safe stopping procedure, to prevent unintended starting or moving of the spreader once the blockage has been cleared.



- Pay attention to the surroundings and any other persons and machines moving in the vicinity. Pay attention to the contours of the terrain as well as other hazards, such as slippery surfaces, loose rocks, or branches and trees that could reach the driver's area.
- Make sure to use only an attachment that is in good operating condition. Perform daily inspections and read the instructions regarding service and maintenance in this manual. Never operate the attachment if the hydraulic systems of both the attachment and the loader are not completely intact. Check the condition of the spreader and mixing shafts regularly.
- Do not remove the protective grille in the spreader or modify the attachment in any other way that would affect its safety.
- Make sure that the loader is equipped with necessary safety equipment, and that they are in order. Seat belt must be used. If there are specific risks at the operating area, appropriate safety equipment must be used. Wear a respirator mask when handling dusty materials.
- Also read the safety instructions of the loader from the operator's manual of the loader. Get familiar with the controls of the loader and how to drive it safely, and also note the effects that slippery surfaces pose to the handling characteristics of the loader.



Familiarize yourself with the controls and handling of the loader at a safe area. Pay particular attention to safe stopping of the attachment and the loader. Keep in mind that the weight of the spreader and slippery surfaces affect the manoeuvrability and handling of the loader.



A running spreader or activated hydraulic circuit creates hazards of entanglement and crushing near the spreader. Never try to reach the mixing or spreader shafts manually; the remaining sand in the spreader must be mixed with suitable tools. Removing the protective grille inside the spreader is allowed only when performing maintenance tasks with the spreader hydraulically disconnected. The spreader must be stopped following the safe stopping procedure before going near it.

Safe stopping of the equipment, before going near the spreader



Always stop the spreader following safe stopping procedure before going near the attachment. Safe stopping procedure prevents unintentional movements of the spreader when filling or when clearing blockages. Never go under a raised loader boom or the attachment. Note that the loader boom can move down even if the engine of the loader is turned off. Safe stopping procedure:

- Stop the attachment and the loader.
- Lower the boom and the attachment on the ground.
- Shut down the loader engine and lock the parking brake.
- Release residual pressure from the hydraulic system.
- Prevent starting of the machine, remove key from ignition.

Remember to wear proper protective clothing:

- The noise level at the driver's seat may exceed 85 dB(A) depending on use and operating conditions. Wear hearing protection while working with the loader.



- Wear safety boots while working with the loader.



- Wear protective gloves.



- When handling hydraulic hoses or components, wear also safety goggles.



- Do not wear loose clothing, scarves, or jewelry, that may get entangled with the rotating parts of the spreader.

4. Technical specifications

Table 2 – Rear mount sand spreader specifications

Product number:	A21242
Working width:	1030 mm
Total width:	1280 mm
Weight:	225 kg
Volume:	280 l
Maximum input of hydraulic energy:	40 l/min, 22,5 MPa (225 bar)
Air pressure in support wheels:	2,0-2,5 bar
Suitable AVANT loader models:	See Table 1
Dimensions:	Page 2

4.1. Safety labels and main components of the Sand Spreader

Listed below are the labels and markings, which must be visible on the equipment. Replace the warning labels, if they have become unclear or if they have detached completely. New labels are available via your retailer or contact information provided on the cover.



The warning labels contain important safety information and they help to identify and remember the hazards related to the equipment. Replace damaged or missing warning labels with new ones.



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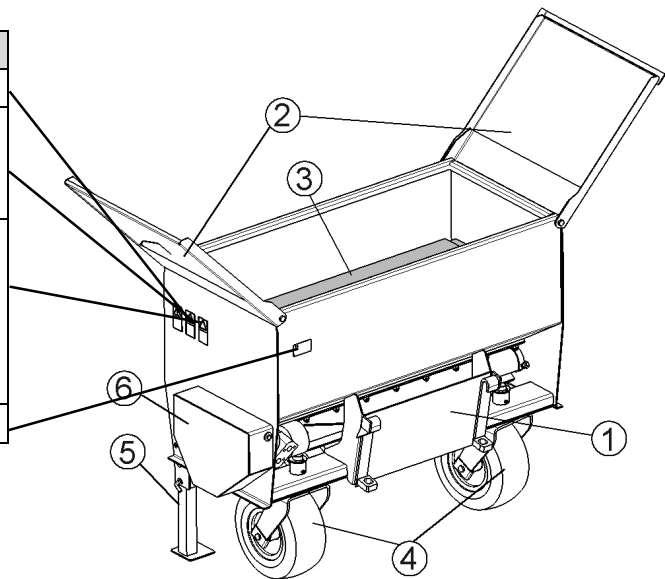
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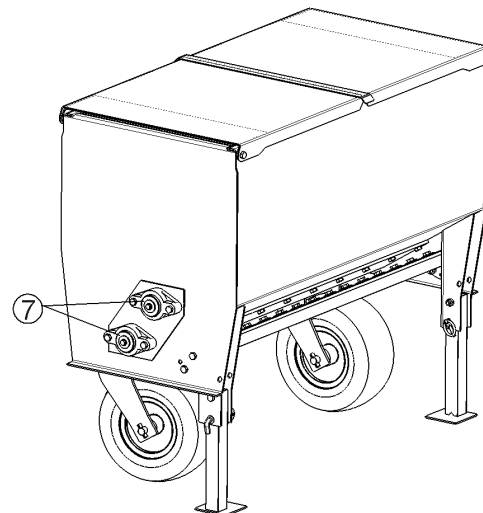
A46801

Table 3 – Decal warning messages and locations

Decal	Explanation
A46771	Read instructions before use.
A46772	Do not go under a raised attachment or loader boom; stay away from raised equipment.
A46801	Entanglement hazard, keep clear of rotating parts. Stop the attachment before leaving the driver's seat. Make sure protective guard (3) and drive chain cover (6) are fitted.
	Attachment identification plates

**Table 4 – Main components of the sand spreader**

1	Quick coupling plate
2	Cover doors
3	Safety grille to prevent reaching to rotating parts
4	Support wheels
5	Outriggers
6	Drive chain cover
7	Bearings of spreading and mixing shafts



 			
Ylötie 1 FIN-33470 YLOJÄRVI			
TYPE	Axxxxx	No.	00000
Year	2010	kW	kg

Identification plate of the attachment

5. Assembly of the sand spreader

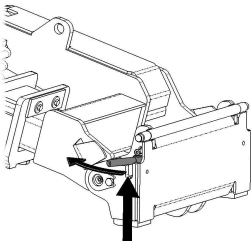
The sand spreader is ready for use as delivered. Attach the attachment and its hydraulic hoses to the loader as instructed below.

If using the spreader as mounted to the back of the loader, refer to instructions about using the rear lift device and rear auxiliary hydraulics outlet from the operator's manual of the loader. The same overall mounting procedure shown below must be followed.

5.1. Attaching the sand spreader to the loader

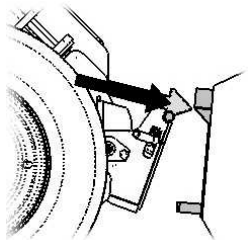
Mounting the attachment to the loader is quick and easy, but it must be done carefully. An attachment that has not been locked may come off from the loader and cause hazardous situations. The attachment is mounted to the loader boom by using the quick attach plate and the counterpart on the attachment. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised to a height of over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3 and the instructions provided in the operator's manual of the loader.

Step 1:



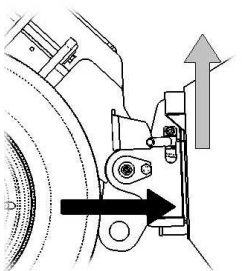
- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.

Step 2:



- Turn the quick attach plate hydraulically to an obliquely forward position (manual tilt on rear lift device).
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this. Align the upper pins of the loader's quick coupling plate so that they are under the corresponding brackets of the attachment.

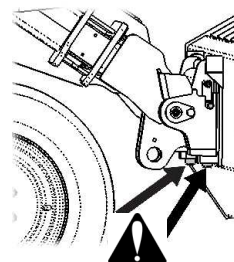
Step 3:



- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment (tilt rear lift plate manually).
- Lock the locking pins manually or lock the hydraulic locking.
- Always ensure positive locking of pins.



An attachment that has not been completely locked to the loader may fall on the boom or operator, or under the loader during driving, causing loss of control of the loader. Before moving or lifting the attachment, make sure that the locking pins are in the lower position and come through the fasteners on both sides. Never move or lift an attachment that has not been fully locked.



5.2. Mounting spreader to rear lift device or rear mount bracket

Follow the same steps as shown previously in Chapter 5.1. Ensure that the spreader will remain stable when mounting it. When mounting to rear mount bracket, suitable lifting straps and equipment must be used.



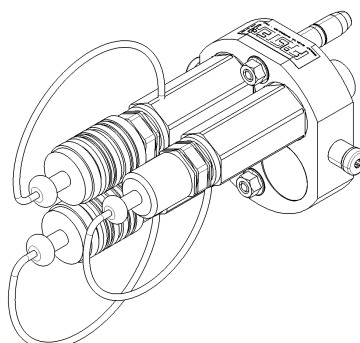
Ensure that a rear mounted spreader does not cause the front of the loader to become too light, so that the front wheels could lift from the ground. Mount a bucket or heavier attachment to the front of the loader for safe and efficient operation.

5.3. Connecting and disconnecting hydraulic fittings

The rear mount spreader is equipped with separate quick couplings for connecting to rear outlet of loader. By using the multiconnector adapter A414508 the spreader can be easily mounted to the loader boom allowing temporary use. If you wish to change the type of the couplings, contact your Avant dealer or service point for instructions or installation service.



Conventional quick couplings



Multiconnector adapter A414508

NOTICE

Keep all fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Conventional quick couplings:

Before connecting or disconnecting the standard quick couplings, the residual pressure must be released as shown below. The conventional quick couplings will not connect, if there is pressure in the hydraulic system.

To connect or to disconnect the standard quick couplings, move the collar at the end of the female fitting. The hoses should be connected so that the fitting equipped with a coloured cap is connected to the corresponding fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious burns or other injuries.

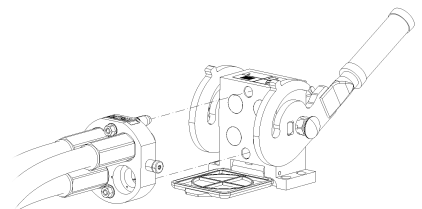
NOTICE

Make sure that all three quick coupling fittings are properly connected, otherwise the hydraulic motor of the attachment may get damaged.

Connecting or disconnecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.



To disconnect the multiconnector system, turn off the auxiliary hydraulics of the loader, and unlock the multiconnector. To unlock, push the knob on the side of the connector, and turn the lever to disconnect.

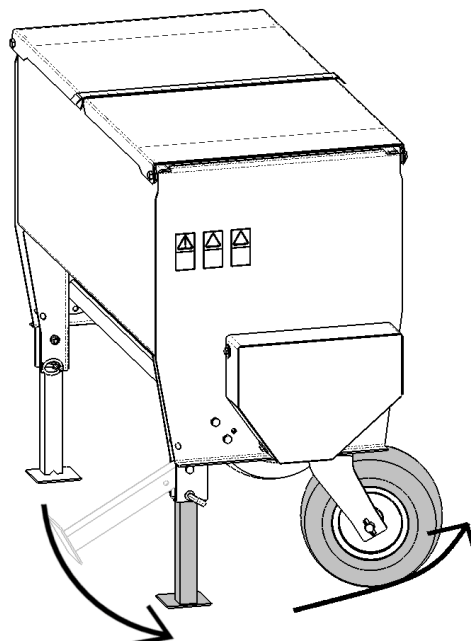
Releasing residual hydraulic pressure:

To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader to its extreme positions before disconnecting the couplings. If the fittings will not connect to the loader, the residual pressure must be released. The residual pressure must be released before handling the rotating components of the attachment and before storage of the attachment.

5.4. Uncoupling and storage of the sand spreader

Uncouple and store the spreader on level surface, so that it will stand firmly on its two outriggers and support wheels. Lower the attachment so that the support wheels touch the ground lightly, and drive the loader to turn the support wheels on the side of the quick coupling plate (or turn wheels manually). Set and lock the outriggers into storage position. Make sure that the spreader will stay stable when uncoupling it and during storage.

Storing the spreader with material inside it may cause the material to stick together, leading to poor performance. Close the spreader doors for storage. The best storage place for the attachment is an area that is protected from rain and direct sunlight, making freezing of sand less likely. Clear most snow from the top of the spreader, so that melting and refreezing snow will not clog the spreader. Climbing on the spreader is prohibited. Service the spreader before longer storage periods, read instructions concerning maintenance and service from Chapter 7.



NOTICE

When uncoupling the attachment from the loader, always disconnect the hydraulic fittings first before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent dirt from entering the hydraulic system. Set the hoses on the bracket on the attachment, or onto the top of the attachment; never leave hoses lying against the ground.



An unmounted spreader may tip over. Make sure that outriggers and support wheels are set properly for storage. Keep a safe distance to the spreader and never climb on the spreader. For the best stability, it is recommended to use the spreader until it is empty, or to empty it manually before storing it.



6. Instructions for use

Check the attachment and the operating area once more before starting work. Quick inspection of the equipment before starting to use it is part of ensuring safety and the best performance of the equipment. Practise the use of the spreader at an open and safe area. If you are unfamiliar with the loader model in use, it is recommended to practice its use without any attachments.



Beware of hazard of entanglement near the rotating parts of the spreader. Never leave the driver's seat, if the auxiliary hydraulics control lever is locked on, or if the spreader or its parts can move unintentionally. Stop the spreader and the loader following the safe stopping procedure before going near the attachment. Always use suitable tools to clear blockages or clogs, never remove the protective grille from inside the spreader, and never try to reach below the grille or spreader shafts manually. Make sure the grille is in its place inside the spreader.



6.1. Checks before use

- Check the general condition of the attachment and the loader, and check for possible hydraulic oil leaks. The attachment must not be used, if there is a fault in the hydraulic system of the loader or the attachment. Refer to Chapter 7 for maintenance instructions.
- Check the condition of the mixing and spreading shafts, and that they can rotate freely. Check that there is no string or other foreign material coiled around them.
- Check that the drive chain cover is securely in its place on the side of the spreader. Using the spreader without the guard in its place is prohibited. Check the drive chain if the spreader makes loud noise during operation.
- Lift and lock the outriggers to upper position always when using the spreader.
- Before starting work, remove larger obstacles from the working area. Remember correct working methods and avoid leaving the driver's seat unnecessarily.
- Drive carefully on uneven terrain and note the effect of heavy load at the rear or at the front of the loader to the stability and handling of the loader.
- Ensure that the loader and the attachment are used in a safe and intended manner. Operate the controls of the loader only when sitting on the driver's seat. Do not allow children to operate the machine.



Check before use that the drive chain protective cover is fastened in its place. Operating the spreader is prohibited, if all covers are not in place.



6.2. Operating the spreader

Filling the spreader:

Fill the spreader only when it is firmly attached to the loader. Lower the spreader on the ground and observe that the spreader and loader might tilt when adding load to spreader. Avoid staying directly in front of the spreader. Stop the spreader before going near it.

The spreaders work the best with crushed rock or gravel that consists of homogenous max. 6 mm grains. Packed or frozen sand or gravel can clog the spreader regardless of the mixing shaft. When finished sanding, do not leave material in the spreader, as the material may clod or freeze and using the spreader the next time may be more difficult.

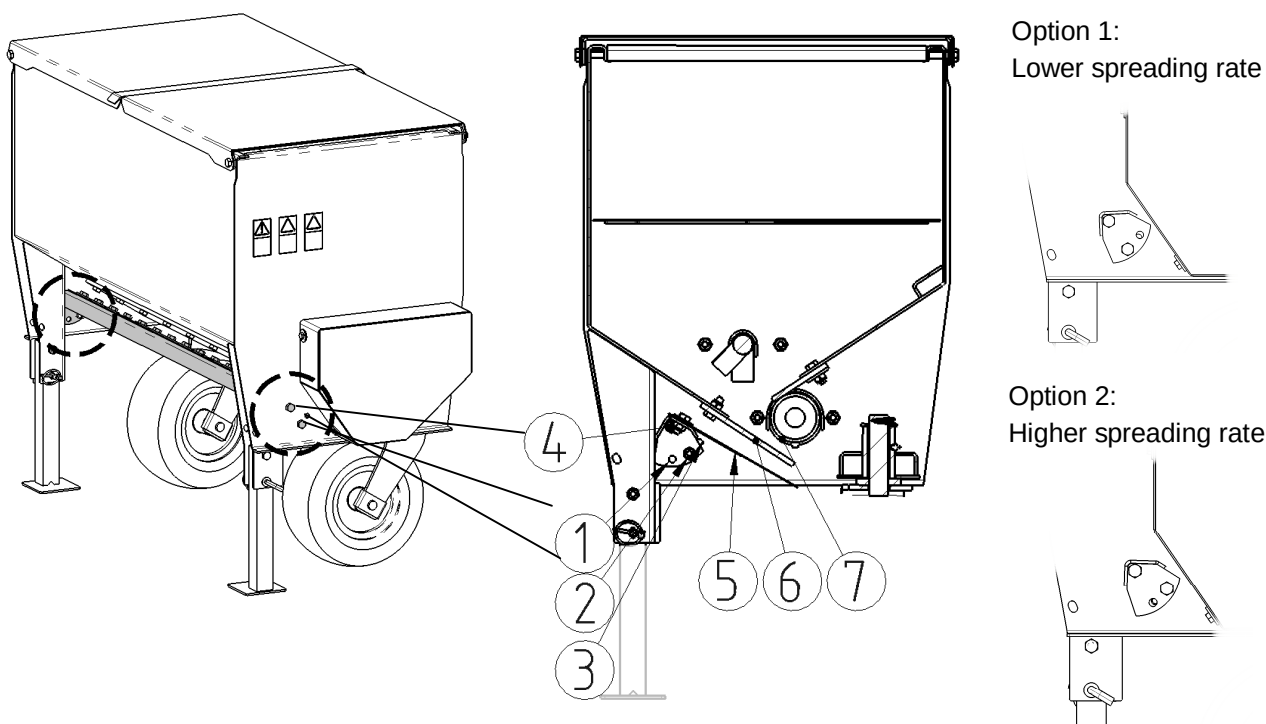
Starting and operating the spreader:

Start the spreader by turning the auxiliary hydraulics control lever of the loader towards its locking position, while the loader engine is set to low rpm. The auxiliary hydraulics control lever or switch is indicated in the operator's manual of the loader. Keep the spreader in upright position near the ground surface, keeping in mind that the support wheels must not touch the ground continuously. Generally, the spreader should be operated using the 1-pump setting of the loader auxiliary hydraulics.

6.3. Adjusting the spreader

The spreading rate depends on the type of material, loader engine rpm, and setting of the spreader. There is a setting to adjust the spreading rate according to most commonly used sanding material and driving style. The spreading function is designed so that a rubber plate (6) is pressed against spreader shaft (7) by a spring plate assembly (5).

The general spreading rate of the spreader is adjusted by changing the position of the spring plate assembly (5), by changing the position of its fastening bolts (1 and 2). Notice that removing the fastening bolts will cause the sand to drain from the spreader, so the adjustment should be made to an empty spreader. There are two options for the setting according to the position of the setting flanges. The smaller spreading rate (option 1) is set when the spring plate assembly is tightened more against the spreader shaft. Make sure that the spring assembly plate will not spring away from the machine, stay at the side of the spreader when removing fastening bolts.



6.4. Clogging of spreader

Freezing or arching of the material inside the spreader can lead to poor spreading performance, but complete blocking is unlikely to occur when using normal spreadable materials. The material inside the spreader can be mixed with a shovel, which usually is enough to clear any blockage. Follow safe stopping procedure before going near the spreader. Keeping the material protected from rain is important for efficient operation. Mixing material with salt may reduce or prevent freezing. Never remove the protective grille from the spreader, always use suitable tools. Also, do not hit the spreader against the ground to prevent damaging the spreader.

7. Maintenance and service

Continuous maintenance of the spreader includes regular cleaning, monitoring the condition of rotating parts and regular lubrication. Check the material left in the spreader and remove any rocks or clogged material. If the spreader makes unusual noise, stop the spreader, empty it and check the mixing and spreader shafts as well as the chain drive.



Ensure that the attachment is properly supported during all inspections, adjustments or maintenance work. Ensure the support also when the spreader is mounted to the loader; note that the loader boom and the rear lift device may move even if the engine is not running.

7.1. Checking hydraulic components

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Do not use the equipment, if you have discovered a leak in the hydraulic system of the attachment or the loader. Leaking hydraulic fluid may penetrate skin and cause serious injuries. Seek medical attention immediately in case hydraulic fluid penetrates the skin. Wash any part of body that has been in contact with hydraulic oil carefully with water and soap. Hydraulic fluid is also harmful to the environment and any leak to the environment must be prevented. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Operate the attachment only with type of hydraulic oil that is accepted for use in Avant loaders.



Never handle hydraulic components when the hydraulic system is pressurized, since a fitting may break or become loose and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



Check hoses visually for cracks or abrasions. If there are signs of leaks, to check a component, hold up a piece of cardboard in the area where a leak is suspected. Do not use hands to search for leaks. Monitor the wearing of the hoses and stop the use, if the surface layer of any hose has worn off. Check the routing of the hoses; adjust the hose clamps to avoid abrasion to the hoses. The hoses have limited service life, which is marked on the hoses. After the date marked on the hoses, they should be replaced with new ones.

Finding any of these faults means that the hydraulic hose or component must be replaced. Spare parts are available from your nearest AVANT retailer or authorized service point. Leave the repair work to professional service technicians, if you don't have adequate knowledge and experience about hydraulic assemblies and how to perform the repairing safely.

7.2. Cleaning and lubrication

The spreader should be cleaned regularly to prevent accumulating dust from freezing or clogging the machine. If ice builds up heavily preventing the normal use of the blower, defrost the attachment in a warm area, as using chemicals may cause damage to the hydraulic hoses. You may use a pressure washer and mild detergent to wash the spreader. Avoid spraying directly at the electric or hydraulic components or using solvents when washing. In addition, mind the labels on the attachment. Lubricate the lubrication point on the bearings after washing.

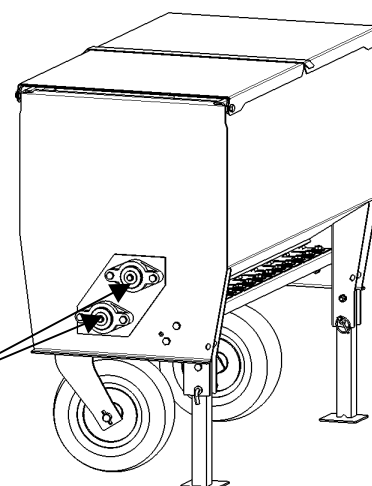
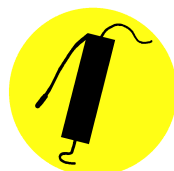
Clean the attachment before long storage periods, touch up the paint where needed and grease the lubrication point to prevent rust damage. Service the drive chain as shown in Chapter 7.3. Ensure that the attachment is supported during maintenance.

Lubrication:

There are lubrication points at the bearings of the mixing and spreading shafts. Additionally, there are lubrication points on the vertical axles of the support wheels, which should be lubricated every 20 hours of use. Clean the end of the nipple before greasing. Only add the grease an amount of a couple of grease press pushes at a time.

NOTICE

Lubricant must be added approximately every 7 to 10 operating hours.



7.3. Drive chain

The hydraulic motor is maintenance free. There is a chain drive between the motor and mixing and spreading shafts. The chain should be lubricated regularly. Remove the protective cover of the drive system and oil the chain after every 20 to 40 hours of operation. Clean the cover and the surroundings of the chain and check the free play of the chain. The chain play should be about 0,5 - 1 cm when measured on its longest run between sprockets. If necessary, adjust the chain by moving the base of the hydraulic motor. Secure the protective cover back to its place after maintenance procedures.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

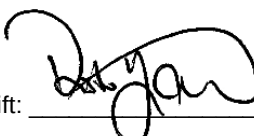
We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:
 SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Kuvaus: Avant-kuormainten kanssa käytettävä hiekoitin
 Beskrivning: Sandspridare; arbetsredskap för Avant lastare
 Description: Sand spreader; attachment for Avant loaders
 Beschreibung: Sandstreuer; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant	300-700-series
Hiekoitin Sandspridare Sand spreader Sandstreuer	A21242

Allekirjoitus / Namnteckning / Signature / Unterschrift:  18.12.2013 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





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